

Network Traffic Analysis with Wireshark

1 - Objective :

The goal of this lab is to analyze a network traffic capture made with Wireshark in order to identify Indicators of Compromise (IOCs) — signals that suggest an attack is in progress or has already occurred.

Based on the IOCs found, possible attack vectors will be identified and concrete actions will be proposed to reduce the impact of the attack.

2 - Our lab enviroment

 **Kali Linux:** Ip 192.168.50.100

VirtualBox Internal Network

3 – Tool Used :

Wireshark Network traffic analysis — capture and inspect packets, protocols, IP addresses and communication content

4 – Results

IOC #1 — Public Service Announcement (BROWSER Protocol)

At row 1, a broadcast packet is observed sent from 192.168.200.150 to 192.168.200.255 via the BROWSER protocol. The machine publicly identifies itself with the name METASPLOITABLE, announcing its exposed services to the entire network.

IOC #2 — Active Port Scan

From row 13 onwards, a very high volume of TCP packets with SYN flag is observed, coming from 192.168.200.100 towards 192.168.200.150, targeting different ports in very rapid sequence. This pattern is consistent with an active port scan of the target machine.

IOC #3 — Vulnerable Samba Service Exposed

Filtering traffic with the "smb" filter, it is observed that machine 192.168.200.150 is broadcasting its Samba version: Samba 3.0.20-Debian. This version is known to contain critical vulnerabilities that allow remote command execution without authentication (CVE-2007-2447).

5 - Issues Encountered During the Test

No major technical issues were encountered during this lab.

The main points that required attention were:

- Identifying the correct Wireshark filter (smb) to isolate the Samba traffic and read the version details
- Recognizing the SYN scan pattern from the raw traffic without any automated tool assistance

6 - Conclusion

The Wireshark traffic analysis revealed clear signs of active reconnaissance on the network. An attacker at 192.168.200.100 mapped the services exposed on the target machine (192.168.200.150) and identified a critical vulnerable service: Samba 3.0.20.

Three IOCs were identified:

- A broadcast packet exposing the machine name and services
- An active SYN port scan
- Samba 3.0.20 exposed on the network, affected by CVE-2007-2447

The identified vulnerability allows full system access without authentication. Immediate action is recommended to reduce the attack surface:

- Update Samba to a non-vulnerable version
- Block SMB ports (139, 445) from external access
- Disable unnecessary network announcements

Technical walkthrough

Step 1 — General Traffic Analysis

At first glance of the capture, an abnormal volume of TCP packets is immediately visible coming from 192.168.200.100 towards 192.168.200.150. The traffic appears very regular and mechanical, with SYN packets sent to different ports in rapid sequence — a typical characteristic of an automated scan.

Step 2 — Syn Traffic Analysis

Observing rows from 13 onwards, TCP packets almost all have the SYN flag active and always come from the same source IP. This pattern is consistent with an active port scan, where the attacker is trying to identify which ports are open on the target machine.

Step 3 — Broadcast Traffic Analysis

At row 1, a BROWSER protocol packet is observed sent in broadcast from 192.168.200.150 to 192.168.200.255. The machine publicly identifies itself with the name METASPLOITABLE and announces its services to the network.

Step 4 — SMB Filter

Applying the "smb" filter in the Wireshark search bar, it was possible to isolate the SMB traffic and analyze the Host Announcement packet in detail. The details panel

clearly shows that the machine is running Samba 3.0.20-Debian — a version affected by critical known vulnerabilities (CVE-2007-2447).

Screenshot

Figure 1 – General Traffic Analysis

No.	Time	Source	Destination	Protocol	Length	Info
1	6.23.76421	192.168.200.1	192.168.200.1	TCP	74	53660 - 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810522427 TSecr=0 WS=128
2	6.23.76426	192.168.200.1	192.168.200.1	TCP	74	53676 - 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810522428 TSecr=0 WS=128
3	6.23.76477	192.168.200.1	192.168.200.1	TCP	74	80 - 53660 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=429495165 TSecr=810522427 WS=64
4	6.23.76477	192.168.200.1	192.168.200.1	TCP	60	53660 - 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810522428 TSecr=429495165
5	6.23.76481	192.168.200.1	192.168.200.1	TCP	60	53660 - 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810522428 TSecr=429495165
6	8.28.76162	PCSSystemtec	PCSSystemtec	ARP	60	who has 192.168.200.109? Tell 192.168.200.150
7	9.28.76164	PCSSystemtec	PCSSystemtec	ARP	42	192.168.200.109 is at 08:00:27:fd:87:1e
8	16.28.77465	PCSSystemtec	PCSSystemtec	ARP	42	who has 192.168.200.150? Tell 192.168.200.109
9	11.28.77523	PCSSystemtec	PCSSystemtec	ARP	60	192.168.200.150 is at 08:00:27:fd:87:1e
10	12.38.77424	192.168.200.1	192.168.200.1	TCP	74	53660 - 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535437 TSecr=0 WS=128
11	13.38.77421	192.168.200.1	192.168.200.1	TCP	74	536120 - 111 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535437 TSecr=0 WS=128
12	14.38.77423	192.168.200.1	192.168.200.1	TCP	74	536120 - 111 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
13	15.38.77436	192.168.200.1	192.168.200.1	TCP	74	53636 - 554 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
14	16.38.77440	192.168.200.1	192.168.200.1	TCP	74	532536 - 135 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
15	17.38.77453	192.168.200.1	192.168.200.1	TCP	74	46138 - 993 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
16	18.38.77461	192.168.200.1	192.168.200.1	TCP	74	41182 - 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
17	19.38.77468	192.168.200.1	192.168.200.1	TCP	74	23 - 41384 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=4294952460 TSecr=810535437 WS=64
18	20.38.77468	192.168.200.1	192.168.200.1	TCP	74	111 - 56120 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=4294952460 TSecr=810535437 WS=64
19	21.38.77468	192.168.200.1	192.168.200.1	TCP	60	814 - 53181 [ACK] Seq=1 Ack=1 Win=0 Len=0
20	22.38.77468	192.168.200.1	192.168.200.1	TCP	60	554 - 50836 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
21	23.38.77468	192.168.200.1	192.168.200.1	TCP	60	184 - 7236 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
22	24.38.77476	192.168.200.1	192.168.200.1	TCP	60	41384 - 23 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535438 TSecr=4294952460
23	25.38.77471	192.168.200.1	192.168.200.1	TCP	60	56120 - 111 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535438 TSecr=4294952460
24	26.38.77474	192.168.200.1	192.168.200.1	TCP	60	8136 - 41384 [ACK] Seq=1 Ack=1 Win=0 Len=0
25	27.38.77524	192.168.200.1	192.168.200.1	TCP	74	21 - 41182 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=4294952460 TSecr=810535438 WS=64
26	28.38.77517	192.168.200.1	192.168.200.1	TCP	60	41182 - 23 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535438 TSecr=4294952460
27	29.38.77533	192.168.200.1	192.168.200.1	TCP	74	50174 - 111 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
30	30.38.77550	192.168.200.1	192.168.200.1	TCP	74	50836 - 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535439 TSecr=0 WS=128
31	31.38.77552	192.168.200.1	192.168.200.1	TCP	74	53660 - 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535439 TSecr=0 WS=128
32	32.38.77556	192.168.200.1	192.168.200.1	TCP	60	113 - 59174 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
33	33.38.77565	192.168.200.1	192.168.200.1	TCP	60	41384 - 23 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952460
34	34.38.77565	192.168.200.1	192.168.200.1	TCP	60	56120 - 111 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952460
35	35.38.77574	192.168.200.1	192.168.200.1	TCP	60	8136 - 41384 [ACK] Seq=1 Ack=1 Win=0 Len=0
36	36.38.77579	192.168.200.1	192.168.200.1	TCP	74	80 - 53662 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=4294952460 TSecr=810535439 WS=64
37	37.38.77580	192.168.200.1	192.168.200.1	TCP	60	59556 - 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952460
38	38.38.77581	192.168.200.1	192.168.200.1	TCP	60	53662 - 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952460
39	39.38.77586	192.168.200.1	192.168.200.1	TCP	60	41162 - 21 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952460
40	40.38.77597	192.168.200.1	192.168.200.1	TCP	60	59556 - 22 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952460
41	41.38.77600	192.168.200.1	192.168.200.1	TCP	60	53662 - 80 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952460
42	42.38.77617	192.168.200.1	192.168.200.1	TCP	74	50836 - 135 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535439 TSecr=0 WS=128
43	43.38.77623	192.168.200.1	192.168.200.1	TCP	74	54220 - 995 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535439 TSecr=0 WS=128
44	44.38.77633	192.168.200.1	192.168.200.1	TCP	74	34440 - 547 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
45	45.38.77638	192.168.200.1	192.168.200.1	TCP	74	33542 - 545 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
46	46.38.77648	192.168.200.1	192.168.200.1	TCP	74	48114 - 255 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
47	47.38.77645	192.168.200.1	192.168.200.1	TCP	60	110 - 54248 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
48	48.38.77645	192.168.200.1	192.168.200.1	TCP	60	895 - 54220 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
49	49.38.77647	192.168.200.1	192.168.200.1	TCP	74	60900 - 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
50	50.38.77649	192.168.200.1	192.168.200.1	TCP	74	33286 - 143 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
51	51.38.77651	192.168.200.1	192.168.200.1	TCP	74	60832 - 25 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
52	52.38.77656	192.168.200.1	192.168.200.1	TCP	74	40954 - 110 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
53	53.38.77667	192.168.200.1	192.168.200.1	TCP	74	37262 - 53 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
54	54.38.77672	192.168.200.1	192.168.200.1	TCP	74	54586 - 500 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
55	55.38.77681	192.168.200.1	192.168.200.1	TCP	60	587 - 34648 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
56	56.38.77684	192.168.200.1	192.168.200.1	TCP	60	33286 - 143 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
57	57.38.77690	192.168.200.1	192.168.200.1	TCP	74	445 - 33842 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=4294952460 TSecr=810535440 WS=64
58	58.38.77690	192.168.200.1	192.168.200.1	TCP	60	256 - 48614 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
59	59.38.77690	192.168.200.1	192.168.200.1	TCP	60	113 - 49608 [RST, ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=4294952460 TSecr=810535440 WS=64
60	60.38.77690	192.168.200.1	192.168.200.1	TCP	60	143 - 33286 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
61	61.38.77690	192.168.200.1	192.168.200.1	TCP	60	255 - 60832 [RST, ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=4294952460 TSecr=810535440 WS=64
62	62.38.77690	192.168.200.1	192.168.200.1	TCP	60	110 - 49654 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
63	63.38.77690	192.168.200.1	192.168.200.1	TCP	60	53 - 33286 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
64	64.38.77690	192.168.200.1	192.168.200.1	TCP	60	500 - 54800 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
65	65.38.77694	192.168.200.1	192.168.200.1	TCP	60	8136 - 41 [ACK] Seq=1 Ack=1 Win=0 Len=0 TSval=810535440 TSecr=4294952460
66	66.38.77694	192.168.200.1	192.168.200.1	TCP	60	40990 - 139 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535440 TSecr=4294952460
67	67.38.77696	192.168.200.1	192.168.200.1	TCP	60	60832 - 25 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535440 TSecr=4294952460
68	68.38.77698	192.168.200.1	192.168.200.1	TCP	60	37262 - 53 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535440 TSecr=4294952460
69	69.38.77711	192.168.200.1	192.168.200.1	TCP	60	487 - 51534 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
70	70.38.77718	192.168.200.1	192.168.200.1	TCP	74	50900 - 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128

***High volume of TCP SYN packets from 192.168.200.100 to 192.168.200.150**

Figure 2 – Broadcast Packet Analysis

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Wireshark: Packet 1: Capture_03_WL3.pcapng
...
Frame 1: Packet, bytes on wire (2288 bits), 288 bytes captured (2288 bits) on interface eth0, id 0
Ethernet II, Src: PCSSystemtec_fd:87:1e (08:00:27:fd:87:1e), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 192.168.200.150, Dst: 192.168.200.255
User Datagram Protocol, Src Port: 138, Dst Port: 138
NetBIOS Datagram Service
SMB (Server Message Block Protocol), Trans Request (0x25)
SMB Mailslot Protocol
Microsoft Windows Browser Protocol
Command: Host Announcement (0x01)
Update Count: 1
Update Periodicity: 2 minutes
Host Name: METASPLOITABLE
Windows version:
OS Major Version: 4
OS Minor Version: 9
Server Type: 0x00019a03, Workstation, Server, Print, Xenix, NT Workstation, NT Server, Potential Browser
Browser Protocol Major Version: 15
Browser Protocol Minor Version: 1
Signature: 6xaa55
Host Comment: metasploitable server (Samba 3.0.20-Debian)
```

***BROWSER protocol packet — machine identifying itself as METASPLOITABLE**

Figure 3 – Syn Scan Pattern

12 36.77414...	192.168.200.1...	192.168.200.1...	TCP	74 41304 → 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535437 TSecr=0 WS=128
13 36.77421...	192.168.200.1...	192.168.200.1...	TCP	74 56120 → 111 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535437 TSecr=0 WS=128
14 36.77425...	192.168.200.1...	192.168.200.1...	TCP	74 33878 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535437 TSecr=0 WS=128
15 36.77436...	192.168.200.1...	192.168.200.1...	TCP	74 58636 → 554 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
16 36.77440...	192.168.200.1...	192.168.200.1...	TCP	74 52358 → 135 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
17 36.77453...	192.168.200.1...	192.168.200.1...	TCP	74 46138 → 993 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
18 36.77461...	192.168.200.1...	192.168.200.1...	TCP	74 41182 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128

**TCP SYN packets in rapid sequence — active port scan confirmed*

Figure 4 – SMB filter

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.200.1	192.168.200.2	BROW	286	Host Announcement METASPLOITABLE, Workstation, Server, Print Queue Server, Xenix Server
Frame 1: Packet, 286 bytes on wire (2288 bits), 286 bytes captured (2288 bits) on interface eth1, id 0						
Ethernet II, Src: PCSsystemtec-fd:07:1e (08:00:27:fd:07:1e), Dst: Broadcast (ff:ff:ff:ff:ff:ff)						
Internet Protocol Version 4, Src: 192.168.200.150, Dst: 192.168.200.255						
User Datagram Protocol, Src Port: 138, Dst Port: 138						
NetBIOS Datagram Service						
SMB (Server Message Block Protocol), Trans Request (0x25)						
SMB MailSlot Protocol						
Microsoft Windows Browser Protocol						
Command: Host Announcement (0x01)						
Update Count: 1						
Update Periodicity: 2 minutes						
Host Name: METASPLOITABLE						
Windows version:						
OS Major Version: 4						
OS Minor Version: 0						
Server Type: 0x00019a03, Workstation, Server, Print, Xenix, NT Workstation, NT Server, Potential Browser						
Browser Protocol Major Version: 15						
Browser Protocol Minor Version: 1						
Signature: 0xaa05						
Host Comment: metasploitable server (Samba 3.0.20-Debian)						

**SMB traffic isolated — Samba 3.0.20-Debian version clearly visible*